

REFLECTIONS

In this lab, I made an XML file named `student_data.xml`. I typed the students inside the root element, each student has an id, a name, a course, and a year. I also added my own data as the third student, then I used the XML validator website to check my file.

The validator said my file was well-formed, so I had no errors to fix. It was easy to use, and it helped me feel sure that my file was correct. I learned that XML has strict rules, but the rules make the data clean and easy to read.

What are the basic rules of XML you applied in this lab?

The first rule I used is the XML declaration at the top of the file. The second rule is that every opening tag must have a closing tag, like `<name>` and `</name>`.

The third rule is correct nesting. The tags must close in the right order, so the tag holds the `<student>`, `<id>`, `<name>`, `<course>` and `<year>` tags. The fourth rule is that the whole file has only one root element, which is `<students>`.

Why does XML need a single root element?

XML needs a single root element so the file has one clear starting point. All the other elements stay inside it, like branches of a tree because this helps programs read the file in the right order. If there are two roots, the file is not well-formed, and the validator will show an error.

How does XML ensure interoperability?

XML is plain text with a standard format, so many different systems can read it. A program in one language can send XML data to a program in another language, and both can understand it. XML also uses tags that describe the data, so people and computers can easily see what each value means, so, this makes sharing data between different apps and platforms easier.